METRIC

A-A-50625(SH)
9 July 1994

COMMERCIAL ITEM DESCRIPTION

LIQUID FLOW SENSOR

The General Services Administration has authorized the use of this commercial item description in preference to MIL-S-16032

Abstract

This commercial item description covers requirements for liquid flow sensors suitable for use in naval ships. Liquid flow sensors include rotameters, flow switches, flowmeters and vortex shedding devices

Salient characteristics

1 Range Unless otherwise specified, the range shall be in liters per second

2. Accuracy. Accuracy shall be plus or minus 2 percent full scale unless otherwise specified

3. Fluid The type of fluid to be monitored shall be as specified

3 1 Pressure Minimum, maximum and normal pressures, in kilograms per square meter, of the fluid shall be as specified.

3 2 Temperature Minimum, maximum and normal temperatures, in degrees Celsius, of the fluid shall be as specified

3 3 Flow rate. Minimum, maximum and normal flow rates, in kilograms per square meter, shall be as specified

3.4 Fluid viscosity Minimum, maximum and normal fluid viscosity shall be as specified

Beneficial comments, recommendations, additions, deletions, and any other pertinent data which may improve this document should be addressed to Commander, SEA 05Q42, Naval Sea Systems Commmand, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160

AMSC N/A FSC 5930

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3.5 Pressure drop Maximum acceptable pressure drop, in kilogram per square meter, through the sensor shall be as specified

4 Rotameters The following shall be as specified

(a) Port connection - 1/2 inch NPT (no metric equivalent) female or as specified
(b) Type of mounting
    (1) Piping
    (2) Panel

5 Flow switch The following shall be as specified

(a) Type of output
    (1) Solid-state - 75 mA, 24 Vdc
    (2) Relay - 2 A, 115 Vac
(b) Electrical connection - 2 meter cable or as specified
(c) Excitation - 115 Vac , 60 Hz
(d) Range of adjustment
(e) Enclosure
    (1) NEMA 4X for general use
    (2) NEMA 7 for hazardous locations
(f) Port - 1/2 inch NPT (no metric equivalent) female or as specified

6 Flowmeter The following shall be as specified

(a) Accuracy - plus or minus 5 percent or as specified
(b) Connection port - 1/2 inch NPT (no metric equivalent) female or as specified

7 Vortex shedding The following shall be as specified

(a) Piping size
(b) Type of output
    (1) 4 to 20 mA
    (2) 1 to 5 Vdc
(c) Excitation - 115 Vac, 60 Hz
(d) Enclosure
    (1) NEMA 4X for general use
    (2) NEMA 7 for hazardous locations
(e) Vibration compensation unless otherwise specified
(f) Cable entrance - 3/4 inch NPT (no metric equivalent) conduit hub or as specified

8 Vibration Vibration requirements shall be as specified

9 Shock Shock requirements shall be as specified

Quality Assurance Provisions

1 Contractor certification The contractor shall certify and maintain substantiation evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract

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2 Regulatory requirements The offerer/contractor is encouraged to use recovered materials in accordance with Public Law 94-580 to the maximum extent practical

3 Preservation, packaging, packing, labeling and marking The preservation, packaging, packing, labeling and marking shall be as specified

NOTES

1 Acquisition requirements Acquisition documents must specify the following

(a) Range
(b) Accuracy
(c) Fluid
(d) Pressure
(e) Temperature
(f) Flow rate
(g) Fluid viscosity
(h) Pressure drop
(i) Shock and vibration
(j) Type of sensor
    (1) Rotameter
    (2) Flow switch
    (3) Flowmeter
    (4) Vortex shedding
(k) Preservation, packaging, packing, labeling and marking

2 NEMA is available from the National Electrical Manufactures Association, 2101 L Street, N.W., Suite 300, Washington, DC 20037-1580

3 Unless otherwise indicated federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094

Preparing activity
Navy - SH
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